

Jeonghoon Lee

PRODUCT ENGINEER

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Summary

Product engineer, five years at digital dentistry startups — fifth engineer at one, team lead after it was acquired. Built the AI that generates a finished 3D crown from an intraoral scan, then rebuilt the platform, operations, and engineering team around it. Goes where the real bottleneck is, even when it sits in an unfamiliar domain — which is where the job gets interesting. Started in firmware, then learned 3D generative models and browser-based 3D CAD, full-stack web, and LLM-based agents on the job, each while shipping with it.

Work Experience

Gloud

Seoul, South Korea

PRODUCT ENGINEER → TEAM LEAD, FOUNDATION TEAM

Aug. 2024 – Jul. 2026

- Gloud puts resin 3D printers in dental clinics and runs the web service behind them: a clinic submits a case, Gloud designs the restoration in real time, slices it, and prints it on the clinic's own printer — collapsing a two-visit procedure into a single visit.
- Joined through Gloud's acquisition of Steinfeld, promoted to Team Lead within three months, then worked with the CTO to turn around the engineering org and the web platform; later handed the web service to another lead to build the company's web-based AI CAD product.
- **AI CAD Platform** Led a team of three to four on the company's next product bet — a web-based CAD tool that grafts custom AI onto a conventional veneer design workflow, replacing the steps designers spent the most time on — carrying over 3D prosthetics generation work from Steinfeld. Internal designers use it in production for veneer cases, cutting per-case design time by more than half versus third-party CAD (3Shape, Exocad).
- **Ops AI Agent** Built an LLM agent that drafts replies to clinic inquiries for the operations team, retrieving from an internal knowledge base kept current by a daily job that diffs new chat logs and documents against it and routes proposed edits to a human reviewer; operators approve every draft before it is sent, and the team adopted it as a daily tool. Built it hands-on with one engineer under my guidance.
- **Sales AI Agent** Answered a request for additional sales-support headcount by shipping an agent instead — built with an intern on the Gemini API with tool calling — that pulls a clinic's recent order history and past complaints from production data so reps can prepare before a visit.
- **Architecture** Untangled a single codebase in which clinic, lab, and admin users were separated only by env vars and account type — where any change risked all three — into three domain-separated applications, then consolidated them into a monorepo that restored code sharing for AI features without reintroducing coupling; the result was independent deploys, a smaller blast radius, and far less merge contention.
- **Order Operations** Rebuilt the order pipeline — absorbing clinic order peaks, routing cases to in-house designers and external labs for real-time design, then through order review, design review, slicing, and print — so operations ran on 4 people instead of ~10 while the clinic base grew roughly 10x over two years. Order-scoped messaging with read and ownership state replaced a time-ordered chat feed and ad-hoc KakaoTalk threads, and clinics gained direct order-status visibility instead of asking by chat.
- **Payments** Redesigned the payment system around subscriptions, rentals, and prepaid credits, retiring an unversioned scheme whose recurring monthly reconciliation errors forced engineers to patch records directly in SQL; that manual intervention fell to rare exceptions.
- **Printer Control** Designed a Raspberry Pi control module — now in volume production in the company's next-generation 3D printer line — that took printer control in-house from an outsourced firmware vendor and consolidated remote power, print-command dispatch over MQTT, and live camera streaming onto a single board, removing add-on hardware at customer sites; coordinated the hardware team and vendor, and stayed hands-on with the engineer implementing it through testing and rollout.
- **Slicing Pipeline** Researched the approach and guided one engineer through several rounds of rework to land an in-house Python 3D slicing engine that moved slicing into the web workflow, eliminating a manual export-desktop-slice-re-upload round trip, and bound slicing profiles to order type to remove a recurring source of operator error.
- **Team & Delivery** Restructured and rebuilt the engineering team after all six original members departed, hiring 12 engineers — most of them new graduates — and growing product engineering to 15; led the resulting 7-person Foundation Team on the company's longer-horizon bets. Moved releases from batched biweekly deploys to continuous deploy on merge to main, backed by backward-compatibility and testing practices the team could hold to.
- **AI-Assisted Development** Made agent-assisted work a team standard: structured the codebase and conventions (agents.md, shared skills) so agent output stayed within guardrails, set up shared MCP configuration per repository so everyone worked from one environment, and automated code review and documentation.
- **Security** Introduced role-based access control on a platform where authorization had lived only in the frontend, and enforced storage rules server-side so uploaded scans and order attachments were no longer readable across accounts.

Steinfeld

Remote

SOFTWARE ENGINEER

Aug. 2021 – Aug. 2024

- Early engineering hire (#5), built the core AI product from the ground up — the technology behind the company’s acquisition by Gloud — and stayed with it afterward as the last of the founding engineering team.
- **AI Pipeline** Co-architected the pipeline that generates a finished crown from an intraoral scan: locating the prepared tooth, detecting the margin line, generating the crown region, occlusal detail, and body, blending the body back to the margin, reconstructing the mesh, and closing the intaglio into a watertight solid — cutting design from hours to under five minutes.
- **Full-Stack Ownership** Worked across every layer of that pipeline — model training, pre- and post-processing, training infrastructure, serving, the web platform, and the CAD tool — with stages built in parallel, rotated between engineers, and often attempted by competing approaches before one was chosen.
- **Models & Validation** Benchmarked over ten 2D and 3D generative and segmentation models — PointNet++, U-Net and pix2pix, GANs, and SDF-based implicit reconstruction among them — to pick the best fit for each stage, judging output at three levels: model metrics during training, geometric metrics between stages, and expert review of the finished crown.
- **Infrastructure** Replaced a per-run routine of cloning a prepared VM image with a GCP Vertex AI training pipeline, and cut preprocessing of tens of thousands of 3D scans from days to hours on serverless workers. Regenerating a dataset became cheap enough to do on demand, which turned experimentation from one run at a time into several a day.
- **Web Platform** Developed and launched a beta web platform for clients to upload scans, receive AI-generated designs, and refine results in an integrated web-based CAD tool.

GE Appliances

Seongnam, South Korea

FIRMWARE ENGINEER INTERN

Jan. 2021 – Aug. 2021

- Developed C firmware for a new washing machine model through to market launch, including the main–sub-board protocol for an automated detergent dispensing system, remote cycle selection, and cloud connectivity.
- Authored and submitted four patent applications on appliance sensing — lint detection via light sensors, laundry clumping via accelerometers, water-temperature fault detection via humidity sensors, and image-based fabric color identification.

Sensorscall

Atlanta, GA, USA

EMBEDDED SOFTWARE INTERN

May 2019 – Jul. 2019

- Wrote C firmware for a smart home device featured at CES 2019 & 2020, implementing VoIP communication, remote Wi-Fi setup, and ambient-light-driven automatic lighting across five integrated sensors.
- Identified and fixed critical memory leaks that were destabilizing the device in long-running use.

Skills

Programming Languages

Python, TypeScript, C++, C

AI/ML

PyTorch, Vertex AI, Hugging Face, Gemini API (tool calling, RAG)

3D & Vision

Three.js, PyTorch3D, OpenCV, Open3D, Trimesh, PointNet++, U-Net/pix2pix, SDF

Backend & Data

Node.js, Express.js, Firestore, MySQL, Redis, MQTT

Frontend

React, Tailwind CSS

Cloud & DevOps

GCP (Cloud Run, Cloud Tasks, Pub/Sub), Firebase, AWS, Azure, Docker, CI/CD

Education

Georgia Institute of Technology

Online

M.S. IN COMPUTER SCIENCE (OMSCS), IN PROGRESS

Aug. 2024 – Present

- Coursework: Deep Learning, Reinforcement Learning, Machine Learning for Robotics, Machine Learning for Trading, Quantum Computing, Video Game Design.

Georgia Institute of Technology

Atlanta, GA, USA

B.S. IN COMPUTER ENGINEERING, GPA 4.0/4.0

Jan. 2019 – Dec. 2020

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, S.Korea

B.S. IN ELECTRICAL ENGINEERING, GPA 3.95/4.3 (MAJOR 4.2/4.3)

Mar. 2014 – Aug. 2021

- Dual degree program with Georgia Tech; military service Oct. 2015 – Jul. 2017.

Honors

National Scholarship for Science and Engineering, Korea Student Aid Foundation – full tuition and stipend

2014 – 2020

Faculty Honors, Georgia Institute of Technology

Spring 2020

President’s Undergraduate Research Award (PURA), Georgia Tech – \$1,500 stipend, Rewind project

May 2020

Grand Societal Challenge Prize (Team Hummingbird Lead), Cornell Digital Agriculture Hackathon – sponsored by Microsoft & Cargill

Feb 2020

Dean’s List, Georgia Institute of Technology

Spring 2019

Dean’s List, Korea Advanced Institute of Science and Technology (KAIST)

Fall 2018